Analogue electronics: SQA-style questions

National 5 · Electronics & control · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1a) Calculate the current through the resistor.
- I = 12 ÷ 300 = 0.04 A

1b) State the formula used.
- Ohm's law: V = IR

1c) State the unit of current.
- Amperes (amps, A)

2a) Calculate the total resistance.
- R_T = 50 Ω

2b) Calculate the circuit current.
- I = 10 ÷ 50 = 0.2 A

2c) Calculate the voltage across the 30 Ω resistor.
- V = 0.2 × 30 = 6 V

3a) Calculate the total resistance.
- R_T = (100 × 200) ÷ 300 = 66.7 Ω

3b) Calculate the total circuit current.
- I = 12 ÷ 66.7 = 0.18 A

3c) State one characteristic of a parallel circuit.
- Each branch has the full supply voltage across it (or: the branch currents add up to the total)

4a) Calculate the resistance of the parallel section.
- (40 × 60) ÷ 100 = 24 Ω

4b) Calculate the total resistance.
- 10 + 24 = 34 Ω

4c) Calculate the circuit current.
- I = 20 ÷ 34 = 0.59 A

5a) Calculate the total resistance.
- $1/R_T$ = 1/10 + 1/20 + 1/30 = 0.1833
- R_T = 5.45 Ω

5b) Calculate the current through the 10 Ω resistor.
- I = 6 ÷ 10 = 0.6 A

6a) Draw a circuit to give 50 Ω.
- Two 100 Ω resistors in parallel

6b) Prove by calculation that the total resistance is 50 Ω.
- $1/R_T$ = 1/100 + 1/100 = 0.02
- R_T = 50 Ω

7a) Calculate the total resistance.
- $R_{parallel}$ = 15 Ω; R_T = 30 Ω

7b) Calculate the circuit current.
- I = 6 ÷ 30 = 0.2 A

7c) Calculate the voltage across the parallel section.
- V = 0.2 × 15 = 3 V

8a) Calculate the current in milliamps.
- I = 24 ÷ 2000 = 0.012 A = 12 mA

8b) The resistor is replaced by a 1 kΩ resistor. State what happens to the current and explain why.
- The current doubles (to 24 mA)
- because the resistance has halved and I = V ÷ R